STAT 432 Homework 09

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Question 1: Choosing an SVM kernel

Setup. The response is *chd* (n = 462; 34.6% cases, so always predicting 0 gives an error of 0.346). The nine predictors are sbp, tobacco, ldl, adiposity, famhist (1 = Absent, 2 = Present), typea, obesity, alcohol and age. I built five folds by randomly permuting the labels 0–4 repeated across the rows (`numpy.random.default_rng(432)`; fold sizes 93, 93, 92, 92, 92). All 20 candidates use these folds. In each fold, a `StandardScaler` is fit on the four training folds only and then applied to both the training and held-out folds. Models are fit with `sklearn.svm.SVC`, and SE = (SD of the five fold errors)/√5.

Candidate settings (20). Linear: $C \in \{0.01, 0.1, 1\}$. Polynomial degrees 2, 3 and 4, each with $C \in \{0.01, 0.1, 1\}$. Radial: $\gamma \in \{0.01, 0.05\} \times C \in \{0.1, 1, 10\}$, plus $\gamma = 0.2$ with $C \in \{0.1, 1\}$. **Held fixed:** the polynomial kernel is $(\gamma\langle x, x'\rangle + r)^d$ with $\gamma = 1/9$ and $r = \text{coef0} = 1$, which keeps the lower-order terms. The radial kernel is $\exp(-\gamma\|x - x'\|^2)$. There are no class weights, and the default tolerance is used.

Kernel	C	γ	CV error	SE
Poly, d = 2	1	1/9	**0.260**	0.012
Linear	0.1	—	0.264	0.014
Radial	1	0.01	0.264	0.010
Poly, d = 2	0.1	1/9	0.271	0.016
Radial	10	0.01	0.271	0.014
Poly, d = 3	0.1	1/9	0.273	0.017
Linear	1	—	0.273	0.015
Linear	0.01	—	0.279	0.012
Radial	1	0.05	0.281	0.012
Poly, d = 4	0.1	1/9	0.284	0.019
Radial	10	0.05	0.286	0.015
Poly, d = 4	0.01	1/9	0.288	0.018
Poly, d = 3	1	1/9	0.292	0.021
Radial	1	0.2	0.307	0.016
Poly, d = 4	1	1/9	0.316	0.019
Poly, d = 3	0.01	1/9	0.329	0.019
Poly, d = 2	0.01	1/9	0.342	0.017
Radial	0.1	0.01	0.346	0.015
Radial	0.1	0.05	0.346	0.015
Radial	0.1	0.2	0.346	0.015

Table Q1. Mean 5-fold CV misclassification rate for all 20 candidates, sorted from lowest to highest.

Preferred model: degree-2 polynomial, C = 1 (γ = 1/9, coef0 = 1), with CV error 0.260. It has the lowest error of the 20 candidates. Quadratic terms and pairwise interactions help a little, while degrees 3–4 with C = 1 and the radial kernel with γ = 0.2 do worse, which shows variance increasing. The radial fits with C = 0.1 predict class 0 for every observation (error 0.346), so they are underfit. The margin over the linear kernel with C = 0.1 (0.264) is 0.004, well inside one SE (0.012). The linear SVM is therefore a defensible simpler choice under a one-SE rule. As a check, I repeated the CV for the top three settings on 20 other random fold splits. The quadratic kernel had the lowest mean error (0.274, vs. 0.280 for linear and 0.284 for radial) and won 12 of the 20 splits. Every kernel stays near 0.26–0.27, which suggests the boundary is nearly linear with mild curvature.

Question 1 code.

```
import numpy as np, pandas as pd
from sklearn.svm import SVC
from sklearn.preprocessing import StandardScaler

df = pd.read_csv("saheart.csv")
df["famhist"] = df["famhist"].map({"Absent": 1, "Present": 2})
predictors = ["sbp","tobacco","ldl","adiposity","famhist","typea","obesity","alcohol","age"]
X = df[predictors].to_numpy(float)
y = df["chd"].to_numpy(int)
n, p = X.shape

# Five folds built by hand, shared by every candidate
rng = np.random.default_rng(432)
fold = rng.permutation(np.arange(n) % 5)

# 20 candidates. Fixed: polynomial gamma = 1/9, coef0 = 1 (so lower-order terms are kept)
cands = []
for C in [0.01, 0.1, 1]:
    cands.append(("linear", dict(kernel="linear", C=C)))
for d in [2, 3, 4]:
    for C in [0.01, 0.1, 1]:
        cands.append((f"poly d={d}", dict(kernel="poly", degree=d, gamma=1/p, coef0=1, C=C)))
for g in [0.01, 0.05, 0.2]:
    for C in [0.1, 1, 10]:
        if (g, C) == (0.2, 10):  # keep to 20 total
            continue
        cands.append(("radial", dict(kernel="rbf", gamma=g, C=C)))
assert len(cands) <= 20

rows = []
for name, params in cands:
    errs = []
    for k in range(5):
        tr, te = fold != k, fold == k
        sc = StandardScaler().fit(X[tr])          # fitting observations only
        m = SVC(**params).fit(sc.transform(X[tr]), y[tr])
        errs.append(np.mean(m.predict(sc.transform(X[te])) != y[te]))
    errs = np.array(errs)
    rows.append(dict(kernel=name, C=params["C"], gamma=params.get("gamma"),
                     cv_err=errs.mean(), se=errs.std(ddof=1)/np.sqrt(5)))
res = pd.DataFrame(rows).sort_values("cv_err")
print(res.to_string(index=False))
```

Question 2: Support Vector Machine Analysis of Coronary Heart Disease (SAheart)

Data suitability. The response is *chd* (binary), with n = 462, p = 9 numeric predictors (sbp, tobacco, ldl, adiposity, famhist coded 1 = Absent / 2 = Present, typea, obesity, alcohol, age) and no missing values. The classes are moderately imbalanced (160 cases, 34.6%), so the baseline error from always predicting "no CHD" is 0.346. With n ≫ p and all-numeric inputs, an SVM is appropriate. The predictor scales differ widely, so standardization is fit on each fold's training data only.

Kernels, tuning, and evaluation. I compared 18 settings. The linear kernel is the additive baseline. Polynomial kernels of degree 2–4 ($\gamma = 1/9$ and coef0 = 1 fixed) allow global curvature and interactions, and the RBF kernel allows local nonlinear boundaries. $C \in \{0.01, 0.1, 1\}$ for the linear and polynomial kernels, and $C \in \{0.1, 1, 10\} \times \gamma \in \{0.01, 0.1\}$ for the RBF kernel. Every candidate was scored by mean misclassification rate on the same five stratified folds (seed 432). For selection I used a one-SE rule: among settings within one SE of the minimum, choose the simplest (linear < lower polynomial degree < RBF, then smaller C).

Setting	CV error	SE	Setting	CV error	SE
Linear, C = 0.01	0.277	0.018	Poly d = 4, C = 0.01	0.288	0.014
Linear, C = 0.1	0.271	0.010	Poly d = 4, C = 0.1	0.281	0.015
Linear, C = 1	0.271	0.014	Poly d = 4, C = 1	0.318	0.017
Poly d = 2, C = 0.01	0.344	0.003	RBF, C = 0.1, γ = 0.01	0.346	0.001
Poly d = 2, C = 0.1 (selected)	**0.268**	**0.009**	RBF, C = 1, γ = 0.01	0.273	0.012
Poly d = 2, C = 1	0.281	0.010	RBF, C = 10, γ = 0.01	0.268	0.009
Poly d = 3, C = 0.01	0.329	0.012	RBF, C = 0.1, γ = 0.1	0.346	0.001
Poly d = 3, C = 0.1 (min)	0.258	0.012	RBF, C = 1, γ = 0.1	0.290	0.008
Poly d = 3, C = 1	0.305	0.010	RBF, C = 10, γ = 0.1	0.309	0.017

Table 1. Five-fold CV misclassification rate and SE (fold SD/√5) for all 18 candidates.

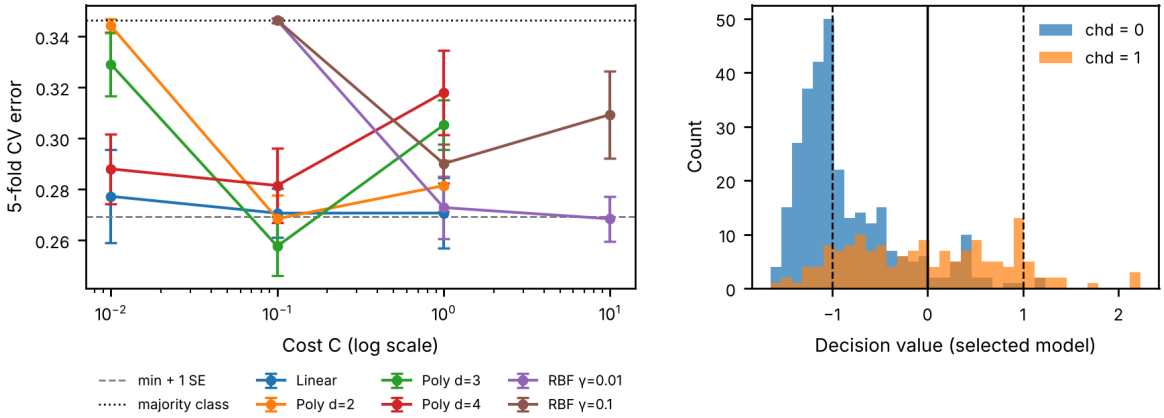


Figure 1. Left: CV error vs. C (±1 SE); dashed = minimum + 1 SE (0.269), dotted = majority-class error. Right: full-data decision values of the selected model by class (solid = boundary, dashed = margins).

Selection. The lowest CV error was the cubic kernel with C = 0.1 (0.258, SE 0.012), giving a one-SE threshold of 0.269. Three settings fall under it: cubic C = 0.1, quadratic C = 0.1 (0.268), and RBF C = 10, γ = 0.01 (0.268). The rule selects the **quadratic kernel, C = 0.1**. The linear kernel with C = 0.1 (0.271) misses the threshold by only 0.001, so the top kernels are not clearly distinguishable. All tuned kernels beat the 0.346 baseline by about 0.08, and very small C collapses to the majority class.

Final fit. Refit on all 462 standardized observations, the model has a training error of 0.242 (CV error 0.268). It uses 293 support vectors (63.4%: 149 non-cases, 144 cases), a wide soft margin that reflects heavily overlapping classes (Figure 1, right). Out-of-fold predictions give TN = 270, FP = 32, FN = 92, TP = 68: sensitivity 0.425 and specificity 0.894. The default boundary therefore misses most CHD cases, so a class-weighted cost or a shifted threshold would be needed if detecting cases matters most. Held-out permutation importance (mean increase in fold error, 30 permutations per predictor, seed 0) ranks famhist (0.047), ldl (0.027), tobacco (0.022), age (0.017) and typea (0.009) highest; the others are ≤ 0.007. In a linear SVM with the same C, family history, tobacco, LDL, age and Type-A score push toward CHD. Obesity has a negative coefficient (−0.24), which likely reflects its collinearity with adiposity rather than a protective effect. **Limitations:** 5-fold CV with n = 462 is noisy (SEs ≈ 0.01), the minimum CV error is optimistic, and no separate test set was held out.

Appendix: SKILL.md

```
---
name: svm-analysis
description: Structured workflow for Support Vector Machine (SVM) suitability assessment,
kernel and hyperparameter tuning, cross-validation model selection, and single-page reporting.
Load only when explicitly invoked.
---
# SVM Analysis

Execute a rigorous Support Vector Machine (SVM) analysis and produce a concise report using
dataset details supplied only at explicit invocation.

## When to Use

- Apply only when explicitly invoked to evaluate, tune, and report an SVM model on a user-
specified dataset.

## Workflow

1. **Assess Data Suitability**: Evaluate sample size, feature count, response type, class
balance, missingness, and numeric encoding of categorical inputs. Require in-fold
standardization because kernel inner products and Euclidean distances are scale-dependent.
2. **Select and Justify Candidate Kernels**: Compare a linear kernel (additive baseline),
polynomial kernels of candidate degrees (global curved boundaries and interactions), and a
radial basis function (RBF) kernel (localized nonlinear boundaries), explaining the geometric
rationale for each.
3. **Identify and Tune Hyperparameters**: Specify regularization cost `C` across all kernels,
degree `d` for polynomial kernels, and scale `gamma` for RBF kernels, while stating any fixed
kernel parameters (such as polynomial scale and offset). Tune across a bounded candidate grid
using shared `K`-fold cross-validation splits with predictors standardized using only the
training observations in each fold.
4. **Evaluate and Select the Final Model**: Score all candidates on the exact same cross-
validation folds using mean out-of-fold error (e.g., misclassification rate) and fold
variability. Select the final model by combining lowest validation error with parsimony
(favoring simpler kernels or stronger regularization when error rates are tied or nearly
indistinguishable).
5. **Include a Compact Justification Table and Figure**: Present one compact table reporting
all candidate settings and mean cross-validation error rates alongside one compact figure
showing cross-validation tuning profiles and model diagnostics.
6. **Interpret the Fit and Report Numerical Results**: Refit the selected specification on the
standardized full dataset and report optimal parameters, cross-validation and training error
rates, support vector count and percentage, class-specific accuracy (confusion matrix,
sensitivity, specificity), and key predictor effects on the decision boundary.
7. **Prioritize Evidence in One Page (11-Point Font)**: Deliver the complete justification,
table, figure, and interpretation in no more than one printed page at 11-point body font,
prioritizing empirical evidence that justifies the final model over secondary commentary.
```